

Daniel Lov

 Daniel Lov  lov-daniel  lovdaniel11@gmail.com  San Diego, California  (626) 636-6294

Objective

Embedded systems engineer with experience in low-level firmware and system software, focused on building efficient, reliable, and resource-constrained embedded systems. My interests include hardware-software co-design, system optimization, and applying academic and project-based experience to embedded systems research.

Experience

Web Developer Intern — *Center for Applied Internet Data Analysis (CAIDA)* **April 2025 - December 2025**

- Developed and maintained responsive user interfaces in React and JavaScript. Enhanced and supplemented core features used by over a dozen researchers, improving performance in load times and API call overhead by 20%.
- Designed and iterated on wireframes and mockups in collaboration with product managers and designers, ensuring alignment between technical feasibility and user experience goals.
- Engineered automation scripts in Python to support product use cases independent of the website interface.

Technical Lead — *Project in a Box, UC San Diego* **June 2025 - Present**

- Conceptualizing and designing hands-on engineering projects to engage students from middle school through college.
- Leading a team of eight to develop a Simon Says-inspired printed circuit board to teach soldering and breadboarding skills.

Course Tutor — *UC San Diego, Computer Science & Engineering* **September 2025 - December 2025**

- Held weekly technical discussions with students to clarify course requirements and expectations.

Undergraduate Researcher — *UC San Diego, Computer Science & Engineering* **September 2024 - June 2025**

- Applied Monte Carlo methods to computational geometry problems, producing scalable solutions in Python with NumPy.

Projects

Pibby Says — *Simon Says PCB Project*  **August 2025 - Present**

- Conceptualizing a Simon Says-inspired interactive electronics project, integrating both Arduino and STM32 microcontroller platforms to handle inputs, outputs, and game logic.
- Currently leading an eight-member team, organizing into specialized sub-teams, coordinating tasks, managing resources, and fostering an environment where members were encouraged to explore new technical skills.
- Designing the PCB layout using KiCad, including schematic capture, netlist generation, component placement, and routing, ensuring manufacturability and functional integrity of the circuit board.

Triton Planning — *Online Degree Planning Platform*  **September 2023 - June 2025**

- Conceptualized and developed a free platform that enables UCSD students to create and share customized degree plans, enhancing the university's standard planning tools.
- Built a responsive single-page application with a React and Bootstrap frontend and a Node.js/Express.js RESTful API backend for secure data processing.
- Implemented MongoDB for efficient data storage and retrieval, integrated Auth0 for secure and seamless user authentication, and deployed the application via Netlify for high availability and performance.

Socket Surge — *Web Sockets Demonstration Workshop*  **September 2024 - May 2025**

- Collaborated in a team of five as the primary backend developer to develop a chess-inspired web application for a workshop designed to teach beginner computer science students how to build real-time applications enabled by WebSockets.
- Designed and implemented the backend with Node.js, Express.js, and WebSockets to enable real-time multiplayer interactions. Developing the frontend with React and Tailwind to provide a responsive and user-friendly interface.

Education

University of California San Diego, B.S. Computer Science (GPA: 3.55)

September 2023 - June 2026

Skills

Languages: Java, JavaScript, Python, C, C++, Bash, SQL, LaTeX, HTML, CSS

Frameworks: React, Node.js, Express.js, Bootstrap, OpenCV2, WebSockets

Developer Tools: Visual Studio, IntelliJ, GitHub, Netlify, Auth0, MongoDB